Q1 Data

3 Points

You will be picking the dataset for this project. Some recommended places to look are:
https://archive.ics.uci.edu/ml/datasets/
https://mavenanalytics.io/data-playground

You must find a dataset that has **at least 10 features**, and enough samples that **each classification problem has at least 200 samples**. For each classification problem, set aside 10% of relevant samples for final validation of the models. This means that you cannot use these samples to train your model parameters, your model hyperparameters, or your feature selection methods.

More details can be found on Canvas

Q1.1 Dataset

1 Point

Your dataset has how many features? How many samples?

16 features with 8000 samples

OR

Upload a file that includes your definitions:

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Q1.2 Variables
2 Points

Briefly describe some background information for your dataset.
What field of knowledge?
What are the variables measuring?
What kind of pre-processing has already been done on your dataset?

EEG (electroencephalogram) signals. The data set was collected from an EEG clinic in Bangalore India. It contains EEG recordings from 80 adult subjects (ages 21–55) with equal gender representation, 20 subjects per category.
The 16 features (X1–X16) represent raw waveform signals from 16 EEG channels corresponding to different brain regions. Each recording lasts 20 seconds. The target variable classifies recordings into four categories: 0 = healthy subjects, 1 = generalized seizures, 2 = focal seizures, 3 = seizure events.
The raw EEG signals have been segmented into fixed 20 second windows and aggregated into 200 time points per subject. The continuous signal values have been converted to integers. The dataset is balanced and no missing values.
According to the introductory paper, initial preprocessing involved exploratory data analysis and data standardization to ensure consistent scales across EEG channels and subjects.

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Q2 Define a real-world motivation

4 Points

You will be fitting classification models to this dataset.

Pick one or two response variables:

The response variable is y, which classifies EEG recordings into four categories: healthy subjects, generalized seizures, focal seizures, and seizure events.

Who would want this model and why? What kind of "action" would they be able to take in order to increase profit, prevent future losses, or gain some other benefit?

Clinicians would want this model to automate the detection and classification of epileptic seizures from EEG recordings. Currently, clinicians manually review of EEG data, which is time-consuming and error prone. With an accurate classification model, clinicians could quickly identify whether a patient is experiencing focal or generalized seizures, which directly determines the treatment plan. The model could also enable real-time seizure monitoring in wearable devices, alerting patients or caregivers before or during a seizure event to take immediate protective action.

Give a reason for training multiple models for the same problem:

Different classification algorithms make different assumptions about the data and capture different types of patterns. For example, the decision tree might capture simple threshold-based rules in the EEG channels, while an SVM with a kernel could find complex nonlinear boundaries between seizure types. Training multiple could help us compare their performance and identify which best handles the specific characteristics of EEG data — such as high dimensionality, temporal dependencies, and class overlap between seizure types. Additionally, if one model overfits while another underfits, comparing them helps find the best bias-variance tradeoff for this particular problem.

OR

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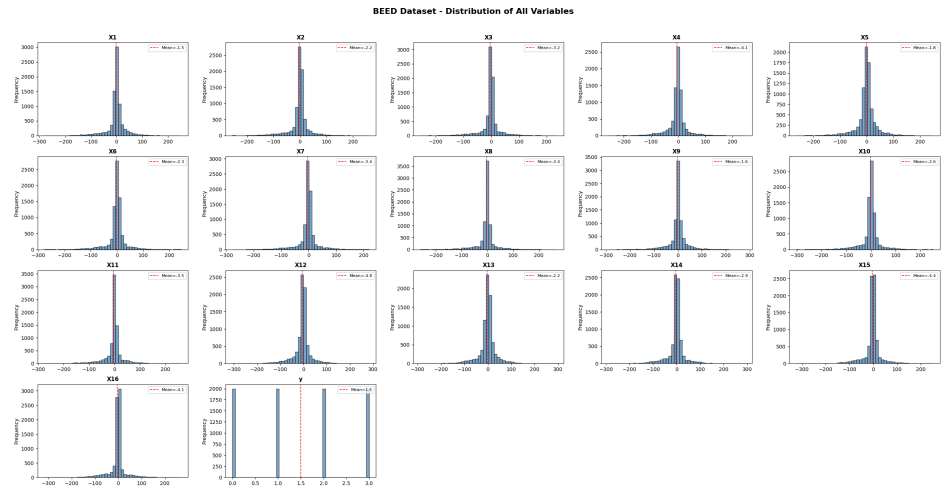
Q3 Visualize variables

2 Points

Create a visualization of the variables and upload them here:

PartAQ3.png Download



Are any of them categorical? normally distributed? Which ones?

Categorical: The target variable y is categorical— it takes only 4 discrete values (0, 1, 2, 3) representing the four seizure classes. All 16 features (X1–X16) are numerical. None of the variables are normally distributed. The features X1–X16 are all slightly left-skewed, they have sharper peaks and heavier tails than a normal distribution.

Q4 Pre-process
4 Points

Process the dataset to create two classification problems.

Q4.1 Pre-process
2 Points

Briefly explain how you defined two distinct classification problems for the same dataset:

Seizure and Healthy (Binary Classification): Remapping the target variable y into two classes: 0 = healthy (original y=0) and 1 = seizure (original y=1, 2, or 3). This creates a clinically relevant binary detection problem that is this EEG recording show any seizure activity or not? This results in 2000 healthy samples and 6000 seizure samples.

4 class seizure type classification (Multiclass Classification): Keeping the original target variable y with all four classes: 0 = healthy, 1 = generalized seizures, 2 = focal seizures, 3 = seizure events. This is important for treatment decisions since focal and generalized seizures require different interventions.

OR

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Q4.2 Datasets
2 Points

Upload your data files here:

BEED_Data.csv Download

1	Large file hidden. You can download it using the button above.

Q5 Model descriptions

4 Points

More details can be found on Canvas.

Reminder:

If you're working by yourself, you only need to pick 2 models.
If you're working with a partner, you'll need to pick 4 models.
If you're working in a group of 3, you'll need to pick 6 models.

Q5.1 Model 1
1 Point

Briefly define/describe your first classification model.

Ex. Linear regression is a model that predicts a numerical response value from a linear combination of numerical input values

k-Nearest Neighbors (kNN) is a classification model that predicts the class of a new sample by finding the k closest training samples in feature space and assigning the majority class among those neighbors. It uses a distance metric (e.g. Euclidean distance) to measure similarity between samples.

Give two strengths for this model by comparing it to an alternative.

Ex. Linear regression is better than ANN because it's easier to interpret the parameters

kNN is better than Naive Bayes because it makes no assumptions about the underlying distribution of the data. Since our EEG features are not normally distributed, Naive Bayes's Gaussian assumption would be violated.

kNN is better than a Decision Tree because it naturally produces smooth, nonlinear decision boundaries. Decision Trees create axis-aligned, blocky boundaries that can miss the complex relationships between EEG channels, whereas kNN can capture more nuanced patterns in the data.

Give two weaknesses for each of your models by comparing it to an alternative.

Ex. Linear regression is unable to explain non-linear relationships, unlike ANN

kNN is slower at prediction time than a Decision Tree because it must compute distances to all training samples for every new prediction.
kNN is more sensitive to irrelevant or noisy features than an SVM, because all features contribute equally to the

distance calculation. If some EEG channels are uninformative, they add noise to the distance metric, whereas an SVM can learn to weight features through its optimization process.

OR

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Q5.2 Remaining models
3 Points

Briefly describe, and give two strengths and two weaknesses, for each of your remaining classification models:

SVM is a classification model that finds the optimal hyperplane that maximizes the margin between classes in feature space. Using the kernel trick it can map data into higher-dimensional spaces to handle non-linearly separable classes.
Strengths:
SVM is better than kNN because it is more robust to irrelevant features. SVM's optimization focuses on the support vectors near the decision boundary, so noisy EEG channels have less impact, whereas kNN treats all features equally in its distance calculation.
SVM is better than a Decision Tree because it generalizes well in high-dimensional spaces. With 16 EEG features, SVM finds smooth decision boundaries that are less prone to overfitting, whereas Decision Trees can create overly complex splits that don't generalize.
Weakness:
SVM is harder to interpret than a Decision Tree. A Decision Tree produces human-readable rules (e.g., "if X3 > 50, then seizure"), whereas SVM's decision boundary that especially with an RBF kernel it is a complex function that doesn't offer intuitive clinical insight.
SVM is slower to train than kNN on large datasets. With 6400 training samples, SVM must solve a quadratic optimization problem, whereas kNN has no training step at all that it simply stores the data.

A Decision Tree is a classification model that recursively splits the feature space based on threshold conditions, forming a tree structure where each internal node tests a feature, branches represent outcomes, and leaf nodes assign class labels.
Strengths:
Decision Trees are more interpretable than SVM because they produce if-then rules. A clinician could directly understand which EEG channels and thresholds are most important for distinguishing seizure types, whereas SVM's decision boundary is not such clear.

Decision Trees are faster at prediction than kNN because classifying a new sample only requires traversing a few nodes in the tree, whereas kNN must compute distances to all training samples.
Weaknesses:
Decision Trees are more prone to overfitting than SVM. A deep tree can memorize noise in the EEG training data by creating highly specific splits, whereas SVM's margin-maximization naturally regularizes the model.
Decision Trees create axis-aligned decision boundaries, making them less flexible than kNN. Since EEG seizure patterns likely involve complex interactions between multiple channels, the rectangular splits of a Decision Tree may miss diagonal or curved class boundaries that kNN captures naturally.

LDA is a supervised classification model that finds linear combinations of features that best separate the classes by maximizing the ratio of between-class variance to within-class variance. It then classifies new samples based on the highest posterior probability.
Strengths:
LDA is more computationally efficient than SVM because it has a closed-form solution that it only needs to compute class means and a shared covariance matrix, whereas SVM must solve an iterative optimization problem.
LDA simultaneously performs dimensionality reduction and classification, unlike kNN which operates in the full 16-dimensional feature space. This makes LDA less susceptible to the curse of dimensionality and noise from uninformative EEG channels.
Weaknesses:
LDA assumes the features are normally distributed within each class, which is violated in our BEED data. kNN makes no such distributional assumptions and can still perform well on non-normal data.
LDA can only learn linear decision boundaries, making it less flexible than SVM with an RBF kernel. If the boundary between seizure types in EEG feature space is nonlinear, LDA will underperform compared to SVM which can capture curved boundaries through kernelization.

OR

Upload your answers here:

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Project 2 Part A

Ungraded

Group

Yuhan Wang
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View or edit group

Total Points

- / 17 pts

Question 1

Data		3 pts
1.1	Dataset	1 pt
1.2	Variables	2 pts

Question 2

Define a real-world motivation	4 pts

Question 3

Visualize variables	2 pts

Question 4

Pre-process		4 pts
4.1	Pre-process	2 pts
4.2	Datasets	2 pts

Question 5

Model descriptions		4 pts
5.1	Model 1	1 pt
5.2	Remaining models	3 pts